

roles of classroom lecture and assignment at home. In the traditional model, the teacher delivers lectures to the students in a classroom. Then the students do their assignments at home, maybe with the help of a parent or Internet. In the flipped classroom model, students are directed to watch short-duration screencasts or video lectures at home before joining the class. Then students do exercises in class in peer or collaborative groups.

Flipped classrooms help to enhance student interactions in the classroom. Flipped classrooms with associated pedagogy of active learning like well-designed think-pair-share, peer instruction, team-pair-solo and paired-problem-solving activity help to enhance problem solving, decision making and conceptualisation skill of learners.

An effective flipped classroom strategy is one in which information transmission happens outside the classroom and information assimilation happens inside the classroom.

Moocs and Moodle. Moodle is an open source web application used to create interactive online learning sites. This free online learning management system enables educators to create their own private website filled with dynamic courses that extend learning anytime, anywhere. Its peer grading feature allows learners to grade and assess their own and other course members' work as a group. Moocs provide free access to organised content with personalised learning environments using blogs, wikis, google groups, Moodle, etc. A large number of free courses are available as Moocs from organisations such as edX.

Discussion forums help with information

TABLE II
MOODLE FEATURES FOR TEACHERS AND STUDENTS

Teachers	Students
Create online courses	Follow online lessons
Upload files and lessons	Upload assignments and tests
Open forums	Can start a thread on the general discussion forum
Divide students into classes	Increase self-directed learning of a topic
Create online tests and quizzes	Perform self-evaluation via quizzes
Provide a platform for students by creating chat sessions	Follow chat sessions in real time
Create wikis and glossary	Work in collaboration

Traditional



Lecture

Homework

Flipped



Video Lecture

In-Class Work

Traditional vs flipped classrooms (Image courtesy: <http://getkindoo.com>)

acquisition, collaboration, concept clarification, review or feedback.

Technology-enhanced learning environment, self-learning, skill improvement and lifelong learning are the key aspects of Moocs and Moodles.

In India NPTEL is a Mooc provider, promoted and funded by the Ministry of Human Resource Development (MHRD), with seven IITs and IISc as its partner institutes. It currently has 160 ongoing courses with about 350 completed courses

and a total of 1.5 million students certified as of July 2017. To promote ICT inclusion, MHRD has also started a national award for teachers using ICT for innovation in education.

Way forward

ICT will become a strong catalyst for a sea change among many educational practices. There is a need to shift from teacher-centered classrooms to student-centric classrooms for outcome-based education. Using technology, teachers can create student-friendly

environments for in-class as well out-of-class learning.

The time is ripe to rethink and redesign the educational systems, processes and ICT-based learning pedagogies. In fact, researchers are working on an amalgamation of cognitive capabilities and the Internet of Things technologies to develop personalised learning tools for both students and educators. With a radical backhaul of the education system in sight, there is also an urgent need to train teachers in the use of ICT tools. **EFY**



NPTEL site visits across countries (Image courtesy: <http://voxitk.com>)